Paleomagnetism of the Deadman's Bay diabase dikes from northeastern central Newfoundland

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Twenty-six oriented samples were collected from six diabase dikes cutting the Deadman's Bay granite of northeastern central Newfoundland for the purpose of paleomagnetic investigations. The dikes probably form part of a general swarm of diabase dikes in this area of Devonian age ($^{40}Ar/^{39}Ar$ age of 370 ± 10 Ma). Stepwise thermal demagnetization of individual specimens followed by vector subtraction based on Zijderveld plots revealed two characteristic magnetizations: one, a poorly grouped magnetization of $\bar{D} = 220.2°$, $\bar{I} = 78.0°$, $\alpha_{95} = 17.4°$, $\kappa = 16$ carried by multidomain magnetite; and a better grouped magnetization of $\bar{D} = 202.9°$, $\bar{I} = 4.0°$, $\alpha_{95} = 10.0°$, $\kappa = 42$ residing in hematite. A baked contact test of the host rock (Deadman's Bay granite) intruded by the diabase dikes revealed a steep well grouped magnetization, which is not dissimilar to that carried by magnetite in the dikes. The magnetization of the host rock seems to be well grouped irrespective of distance away from the dike, thus making the contact test inconclusive. It is not certain which, if any, of the two obtained poles corresponds to the (primary) age of intrusion of the dikes. The paleopole based on magnetite's magnetization, called the DB1 pole, is in the same general vicinity as some "anomalous" pole positions for intrusives from Newfoundland and New Brunswick. The paleopole based on hematite's magnetization, called the DB2 pole, happens to be the first one from eastern Newfoundland that agreed within error limits with the Acadian poles. The poles from diabase dikes of the Gander Zone are discussed in the context of other poles from the northern Appalachians. It is concluded that the Devonian poles from Acadia seem to be displaced relative to the cratonic North American poles for that age. However, critical testing for the confirmation of such a displacement during the Devonian had to wait a clearer definition of the positions of the poles and their ages. It was shown that such a displacement, if present during the Devonian, disappeared during the Mississippian times, suggesting that any relative motion between the two regions ceased by the Mississippian times.

Vingt-six échantillons furent collectionnées à partir de six dykes de diabase recoupant le granite de la baie de Deadman dans la partie nord-est centrale de Terre-Neuve pour une étude paléomagnétique. Les dykes forment probablement une partie d'un réseau général des dykes de diabase d'âge dévonien de cette région (âge de 370 ± 10 Ma par $^{40}Ar/^{39}Ar$). La désaimantation au moyen de traitements thermiques par étapes des spécimens individuels suivie d'une soustraction vectorielle établie avec les diagrammes de Zijderveld a révélé l'existence de deux aimantations distinctes: une, dont les directions d'aimantation sont relativement dispersées avec $\bar{D} = 220{,}2°$, $\bar{I} = 78{,}0°$, $\alpha_{95} = 17{,}4°$, $\kappa = 16$, portée par une magnétite à multidomaines; et l'autre présentant des directions d'aimantation mieux regroupées avec $\bar{D} = 202{,}9°$, $\bar{I} = 4{,}0°$, $\alpha_{95} = 10{,}0°$, $\kappa = 42$, résidant dans l'hématite. Un essai sur la roche-hôtesse, au contact durci par la chaleur due à l'intrusion des dykes de diabase, révèle un groupement bien défini de directions d'aimantation fortement inclinées qui n'est pas dissemblable de celui porté par la magnétite dans les dykes. L'aimantation de la roche-hôtesse semble posséder des directions bien regroupées, quelle que soit la distance au dyke, ce qui empêche de tirer une conclusion de l'essai au contact. Il n'est pas possible de déterminer lequel, s'il y en a un, des deux pôles obtenus, qui correspondrait à l'âge (primaire) de la mise en place des dykes. Le paléopôle déduit de l'aimantation de la magnétite, appelé le pôle DB1, présente la même position générale que les positions de quelques pôles "anormaux" pour les roches intrusives de Terre-Neuve et du Nouveau-Brunswick. Il arrive que le paléopôle déduit de l'aimantation de l'hématite, appelée le pôle DB2, soit le premier dans l'est de Terre-Neuve à concorder, en dedans des erreurs limites, avec le pôle acadien. Les pôles correspondant aux dykes de diabase de la zone de Gander sont interprétés en les comparant avec d'autres pôles des Appalaches du Nord. On arrive à la conclusion que dans l'Acadie, les pôles du Dévonien semblent avoir glissé par rapport aux pôles de même âge du craton de l'Amérique du Nord. Cependant, une démonstration plus éclatante de cette assertion d'un déplacement des pôles durant le Dévonien doit attendre une détermination plus précise des positions des pôles correspondant à ces âges. Il a été établi qu'un tel déplacement, s'il se manifestait au Dévonien, a disparu durant les temps du Mississippien suggérant ainsi que tout mouvement relatif entre ces deux régions n'a pu exister durant le Mississippien.

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Introduction

Paleomagnetic results are reported for a suite of diabase dikes cutting the Deadman's Bay granite of northeastern central Newfoundland. These dikes cut the Deadman's Bay granite west of the village of Deadman's Bay and are prominent in the geological landscape of the area. In the geological literature, not much attention has been given to the significance of the dikes in the tectonic development of the island of Newfoundland. Geochemical data are not available for the dikes. It is possible, however, that they form part of a general diabase dike swarm in northeastern central Newfound-

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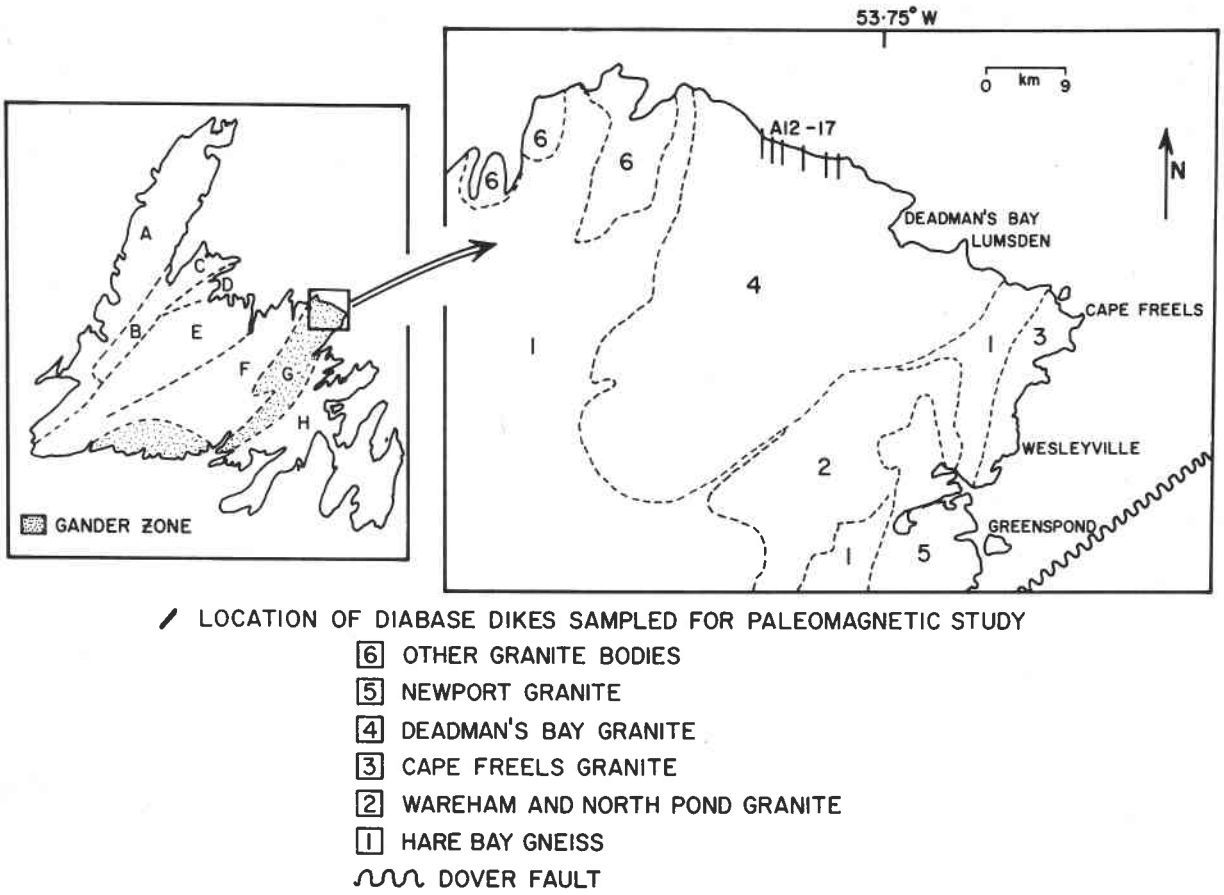


FIG. 1. Generalized geological sketch map of northeastern portion of the Gander Zone showing paleomagnetic sampling localities of the Deadman's Bay dikes (A12–A17). Also shown are the boundaries of various granite plutons in the area and of the Hare Bay gneiss. Numbers 1–6 are from the oldest to the youngest rock unit in the area. The tectono-stratigraphic zoning of Newfoundland (after Williams *et al.* 1972) is shown to the left.

land that was recognized earlier (Jayasinghe 1978) as evidence of a tensional environment in the Devonian. The age of intrusion of this set of dikes is important for later tectonic interpretations of the paleomagnetic results. There were, earlier, partially successful attempts to date radiometrically the Deadman's Bay granite (Berger and Naylor 1974). The age of 510 ± 10 Ma reported was considered preliminary (A. R. Berger, personal communication, 1977) and later attempts by Rb/Sr methods were unsuccessful (Bell *et al.* 1977). One of the diabase dikes collected for paleomagnetic study (see next section) was dated by P. Reynolds by the Ar^{40}/Ar^{39} method. An age of 371 ± 8 Ma was obtained for this dike from both hornblende and a concentrate of biotite, however, with the apparent age plateaus showing a "staircase" pattern. This age is similar to the age of diabase dikes to the east (an age of 373 ± 9 Ma for another dike to the east by Ar^{40}/Ar^{39} method, P. Reynolds and G. S. Murthy, in preparation) and is also consistent with the general cross-cutting relationships farther to the east between the dikes and granites, discussed in Murthy (1983). Thus it is considered likely that the age of intrusion of the Deadman's Bay diabase dikes is 375 ± 10 Ma. In our quest for obtaining reliable paleomagnetic data for eastern Newfoundland, all the diabase dikes in the area were sampled for systematic paleomagnetic studies. In this paper, the paleomagnetic results of the dike swarm west of Deadman's Bay Village are reported. Results of the Bonavista Bay dikes to the east are reported elsewhere along with a comprehensive description of the objective of study of the Gander Zone rocks (Murthy 1983).

Six diabase dikes were sampled (Fig. 1). The dikes, on the average, are 5 m wide and seem to be extremely fresh. Twenty-six oriented cores, each 2.5 cm in diameter and 15 cm long, were drilled in location from the six dikes, with cores uniformly spaced across the width of the dike. Cores were taken from chilled margins as

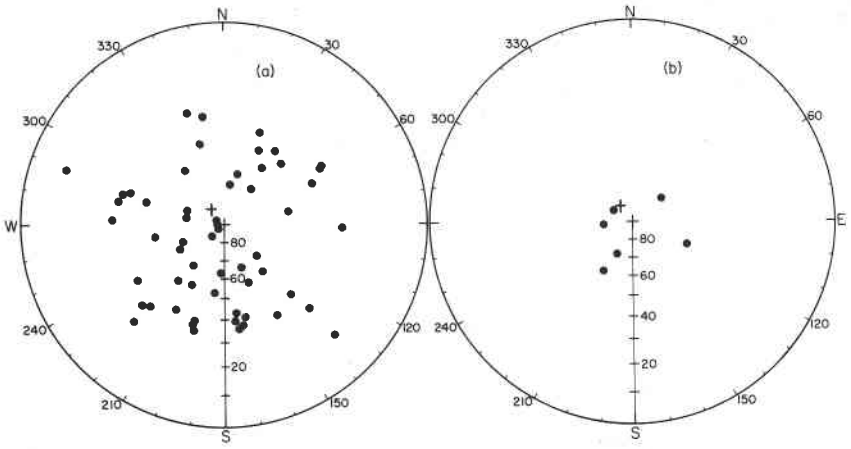


FIG. 2. Natural remanent magnetism of the Deadman's Bay dike samples: (a) specimen directions. † indicates direction of the present Earth's magnetic field. All are positive inclinations plotted on Wulff's stereographic projection.

well and correspondingly marked. Fifty-eight individual specimens were cut from these cores for the paleomagnetic investigations.

Paleomagnetism and rock magnetism

Remanent and bulk magnetic properties of each individual specimen were studied. A Schonstedt SSM-1 spinner magnetometer was used for measuring the natural remanent magnetization (NRM). The bulk magnetic properties studied include the magnetic susceptibility and its variation with temperature, Curie temperatures, tests for the presence of Rayleigh loops, and other hysteresis parameters. All the specimens have high magnetic susceptibilities ($\sim 10^{-2}$ SI units) with the Koenigsberger ratios typically close to one. The specimens did not exhibit any Rayleigh loops. From the absence of Rayleigh loops it is concluded that, although the specimens may have a significant viscous remanence (closeness of the NRM directions to that of the present magnetic field of the Earth and the high magnetic susceptibilities of specimens indicating the *capability* of the specimens to acquire viscous magnetization), it is probably not due to the magnetic grains in the effective superparamagnetic size range. The intensity of the magnetic remanence of the specimens is quite strong with magnetic moments (for a 10 cm^3 specimen) ranging between 4.20×10^{-4} and 2.75×10^{-7} A m^2. The remanence direction of each individual specimen is shown in Fig. 2*a* and the directions averaged for each dike are in Fig. 2*b*. The directions tend to be with steep positive inclinations, close to the direction of the present Earth's field. Alternating field (AF) and thermal demagnetization procedures were used to isolate components of remanence and to see if a characteristic magnetization could be identified. Alternating field demagnetization did not prove to be successful. More than 90% of original intensity was lost in fields of the order of 10 mT and the directional variation was not uniform. Beyond 10 mT, both directional and intensity variations were irregular. During thermal demagnetization, *each* individual specimen was treated in stepwise fashion at the following temperatures: 100, 200, 300, 400, 500, 530, 560, 600, 610, 620, 635, 650, and 680°C. Response of typical specimens is shown in Fig. 3 both as Wulff's net and as vector (Zijderveld 1967) diagrams. The magnetization of each specimen was analysed by either the vector subtraction method based on Zijderveld plots or the stable end-point criterion. As may be noticed in Fig. 3*a*, the remanence direction remained close to that of the NRM up to 560°C and then moved significantly to shallow inclinations for steps beyond 600°C. Stable end points were also achieved beyong the 600°C step (Fig. 3*a*). This behavior suggests that a softer magnetization is carried by magnetite and the harder, finally isolated, magnetization by hematite and that the two magnetic minerals coexist in the specimen. The intensity variation (Fig. 3*b*) suggests that the remanence carried by magnetite constitutes more than 98% of the total remanence. Zijderveld diagrams in Figs. 3 *c* and *d* and 4 show that the demagnetization vector end points fall on a reasonably straight line up to 560°C. Based on these plots, magnetization carried by magnetite was isolated for each specimen by vector subtraction between 200 and 560°C steps. These directions, for individual specimens and combined to obtain dike mean directions, are plotted in Fig. 5*a* and Fig. 5*b*, respectively. The directions carried by hematite are obtained by the stable end-point criterion beyond the 620°C heating step. Zijderveld plots are not used to obtain the hematite's magnetization as the remanent intensity beyond the 600°C heating has fallen to a very minute fraction of the original intensity and any amount

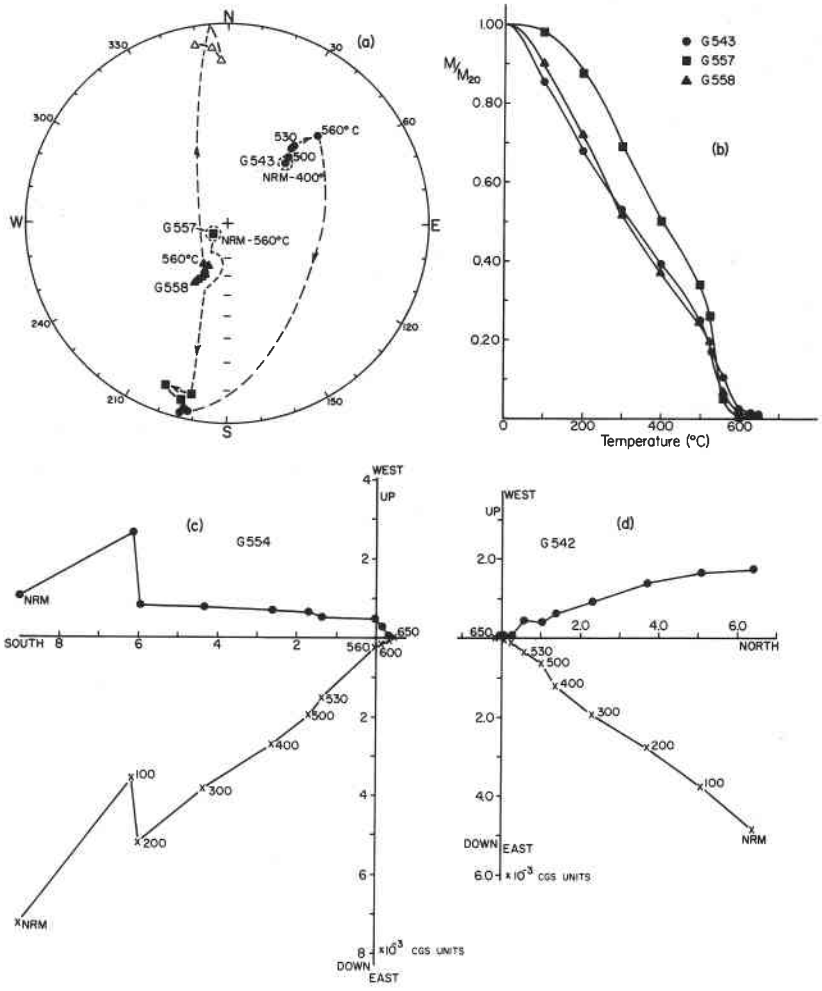


FIG. 3. Response of specimens of Deadman's Bay dikes to stepwise thermal demagnetization. (*a*) Change of direction with heating plotted on a Wulff's net. Solid (open) symbols indicate positive (negative) inclinations. (*b*) Corresponding intensity variations. (*c*), (*d*) Vector (Zijderveld) diagrams of magnetization during thermal demagnetization. Solid symbols are end points projected on the horizontal plane and crosses are end points projected on the north–south vertical plane. Numbers next to crosses are temperature values (in degrees Celsius) used during demagnetization.

of blowup of this tiny fraction can cause inaccuracies. Further, as Dunlop (1979) pointed out, obtaining the final hard magnetization of the specimen by the vector subtraction technique may not be of advantage as the same can be obtained by the stable end-point criterion on a stereonet. The individual specimen directions, and those combined to the dike mean level, are shown in Fig. 5*c* and Fig. 5*d*, respectively.

It is obvious from the shape of the demagnetization curves (Fig. 3*b*) that there is very little hematite in the specimens compared with magnetite. Isothermal remanent magnetization (IRM) acquisition curves were obtained for the specimens in order to see if hematite's presence can be detected by magnetic experiments. Figure 6*a* indicates that the specimen's magnetization saturated in fields of 200 mT, suggesting that there is a predominance of magnetite and further that the presence of any little amount of hematite coexisting with magnetite cannot be deciphered by these experiments. By applying a back field, the coercivity of remanence of the specimens was found to be of the order of 20 mT. Responses to AF demagnetization of the saturation IRM and NRM were compared in Fig. 6*b* in order to infer the domain state of the remanence carriers (Lowrie and Fuller 1971). The IRM is harder to demagnetize than the NRM, suggesting that remanence might be carried by multidomain grains. Further supporting evidence for multidomain state is a remanent coercive force of 20 mT and a median destructive field for IRM of about 8 mT (Dankers 1981). Curie temperatures were also deter-

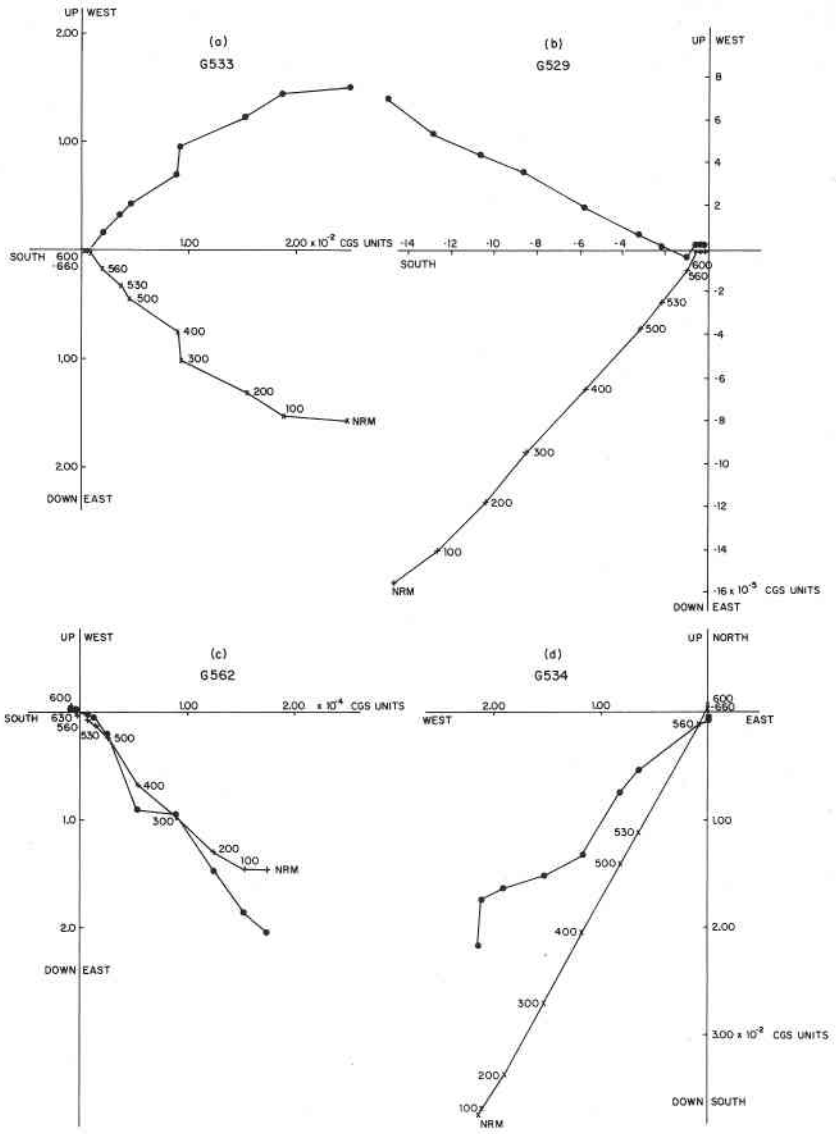


FIG. 4. Vector (Zijderveld) diagrams for specimens during thermal demagnetization. Solid symbols are end points projected on the horizontal plane and crosses are end points projected on the vertical plane. Numbers next to crosses are temperature values (in degrees Celsius) used during demagnetization.

mined for the samples in order to see if presence of hematite can be detected. An apparatus similar to that described by Deutsch *et al.* (1971) was used for this purpose. Figure 6*c* shows that almost all of the magnetic moment was lost by 580°C, and the presence of hematite was not indicated. Hematite, coexisting with magnetite, can only be deciphered by Curie temperature determinations if the amount of hematite is several times larger than the amount of magnetite. The complete reversibility of the curves indicates that no significant change of magnetic minerals took place during heating. Variation of magnetic susceptibility with temperature from liquid nitrogen temperatures to 700°C are shown in Fig. 6*d*. The peak in normalized susceptibility at −155°C and a flat heating curve, followed by a sudden drop in susceptibility at 570°C, are indicative of multidomain bulk behavior of the sample. Hysteresis curves were obtained at room and at liquid nitrogen temperatures (not shown). Doubling of the coercive force at low temperatures is suggestive of dominant multidomain bulk behavior (Radhakrishnamurty *et al.* 1978). Thus all the magnetic experiments to determine the nature of the magnetic mineral and its domain state suggest that magnetite is the predominant magnetic mineral pri-

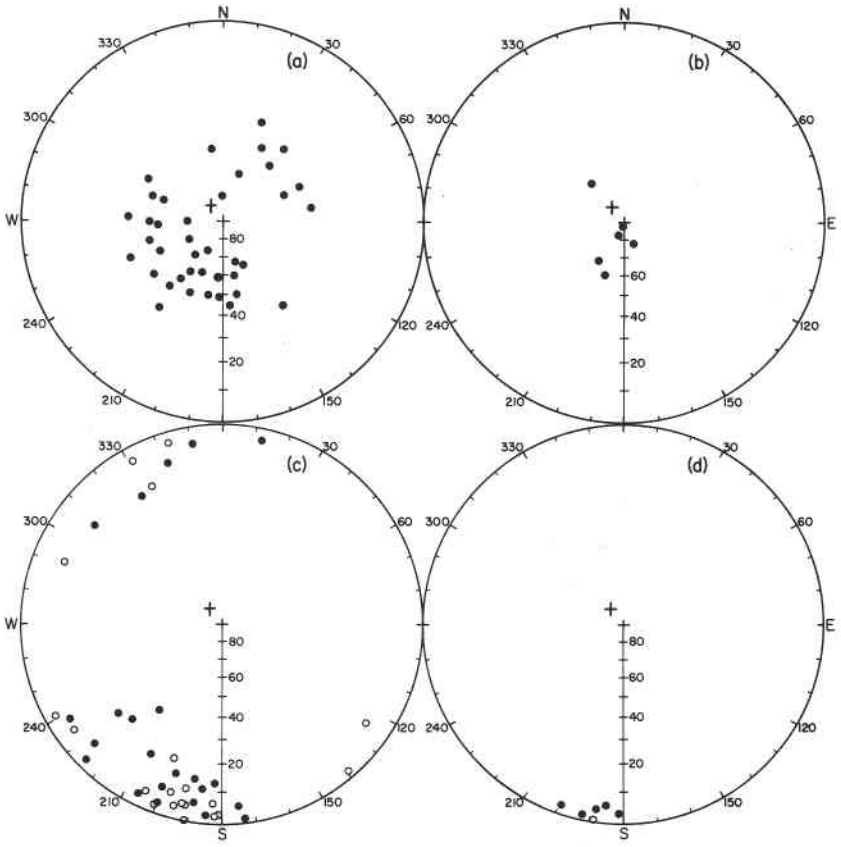


FIG. 5. Characteristic magnetizations for the Deadman's Bay dikes obtained after thermal demagnetization. (*a*), (*b*) Magnetization carried by magnetite isolated by vector subtraction between 200 and 560°C steps based on Zijderveld plots: (*a*) specimen directions; (*b*) dike mean directions. (*c*), (*d*) Magnetization carried by hematite isolated by the stable end-point criterion, usually after the 640°C step: (*c*) specimen directions; (*d*) dike mean directions. Specimen magnetizations with NNW declinations of (*c*) are not included in the calculations of mean directions shown in (*d*).

marily occurring in multidomain form. Thermal demagnetization experiments showed that the harder remanence is carried by hematite and a softer one by multidomain magnetite. Because of hematite's coexistence with magnetite, other magnetic experiments could not decipher the presence of hematite.

Discussion

Almost all the specimens were found to carry two definitely recognizable magnetizations, one carried by magnetite and the other by hematite. These directions are plotted in Fig. 5 *a* and *c* and the corresponding dike mean directions in Fig. 5 *b* and *d*. The magnetization carried by magnetite is rather dispersed. Although part of this dispersion is attributable to the method used in isolating this magnetization, this dispersion is also probably reflecting the complex nature of the dike magnetization. A large number of magnetite magnetizations have a direction that is intermediate between the hematite direction and the direction of the present-day Earth's field. This suggests that the present-day magnetic field may be one of the three or four possible magnetizations recognizable in the total magnetization of the specimens. Figure 5*a* seems to indicate that two groups of magnetization carried by magnetite are isolated: 10 specimens with northeasterly declination and 29 specimens with southwesterly declination. Although the results of an *F*-test show that the two groups are distinct, there is neither geological nor physical reason to believe that two distinct magnetizations of magnetite are isolated. This also contributes significantly to the dispersion in the mean magnetization for magnetite quoted in Table 1. Magnetite's directions are similar to the NRM directions, which confirms that magnetite's magnetization dominates the natural remanence. A paleopole, called the DB1 pole, was calculated from magnetite's magnetization and is lised in Table 1.

Characteristic magnetization residing in hematite is with shallow inclination (Fig. 5*c*). However, the specimens are found to have two distinct declination group-

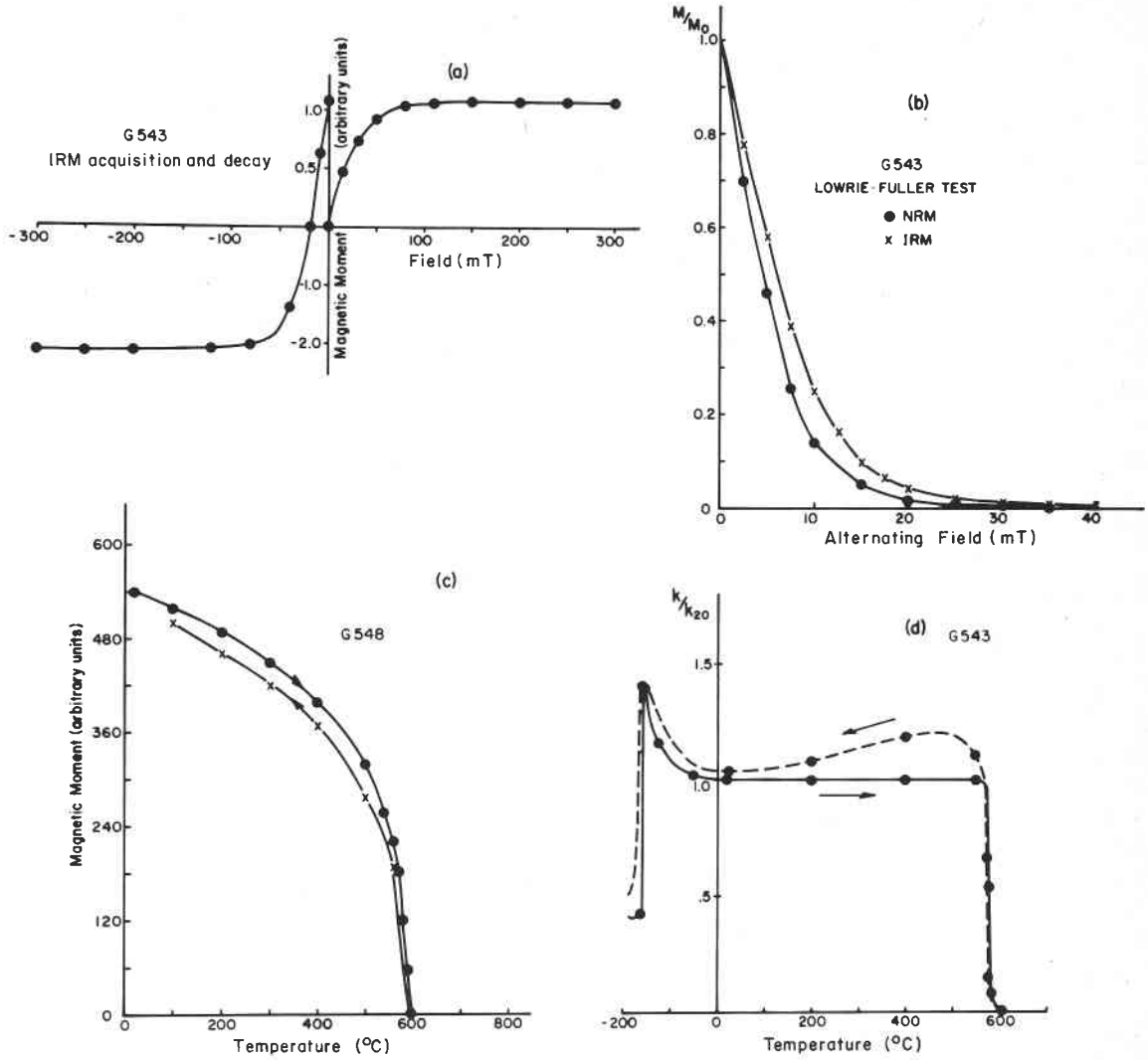


FIG. 6. (*a*) Acquisition of IRM and its decay on application of back field. (*b*) Response of NRM and of IRM of specimen G543 to alternating field demagnetization (Lowrie–Fuller) test. (*c*) Magnetic moment (arbitrary units) versus temperature in a field of 860 mT, heating and cooling. (*d*) Normalized susceptibility versus temperature curve for specimen G543 in a field of 31 μT, heating and cooling.

ings: the majority are in south-southwesterly direction and there is a minor grouping north-northwesterly. The two groupings are not 180° to each other: the second grouping has a bias in declination to the west. An *F*-test was conducted to check if we are dealing with one magnetization in the two declination groupings. The test resulted in a large value of the *F*-statistic suggesting that the assumption of a single grouping is false. Thus it is not statistically meaningful to calculate a mean magnetization carried by hematite in all the specimens. Of the two magnetizations carried by hematite, it is probable that the one with south-southwesterly (SSW) declination may be more representative of a paleomagnetic field, being carried by the majority of specimens. The origin of the north-northwesterly (NNW) magnetization is unclear. It is possible that this magnetization represents a different time compared with the magnetization with the SSW declination. Specimens with the NNW declination are uniformly distributed from all the dikes except A12. There is no preferential location of these specimens from either the edge or middle of the dikes. However, if a hydrothermal event had happened in the area affecting the weakest zones within each dike, thus causing selective remagnetizations, then it is likely that the NNW magnetization in hematite may be representative of geomagnetic field at a later time. Thus, rather

TABLE 1. Characteristic magnetizations of the Deadman's Bay dikes

Dike	NRM				Magnetite's magnetization[a]				Hematite's magnetization[b,c]				
	N	D (°)	I (°)	α_{95} (°)	κ	n	D (°)	I (°)	α_{95} (°)	κ	n	D (°)	I (°)	α_{95} (°)	κ
A12	4	53.3	69.5	1.9	157	7	320.1	60.1	37	3.6	7	198.7	2.8	21	9.0
A13	5	264.7	73.3	3.2	51	6	206.6	87.3	32	5.4	5	195.6	−1.6	9.8	62
A14	4	301.2	78.2	4.9	47	6	206.4	80.9	43	3.4	6	200.1	4.9	18	14
A15	5	211.6	58.6	22	20	8	200.1	57.6	13	18	5	217.4	8.9	25	10
A16	3	205.3	69.8	9.1	24	5	214.6	63.0	30	7.6	3	217.7	4.5	24	28
A17	5	113.1	57.3	2.4	36	7	158.0	76.0	40	3.2	6	188.1	3.8	9.7	49
Mean	6	188.7	82.9	21	11	6	220.2	78.0	17	16	6	202.9	4.0	10	42
								Pole position, DB1					Pole position, DB2		
								Lat. = 30.1°N					Lat. = 35.0°S		
								Long. = 70.7°W					Long. = 82.1°W		
								(δ_p, δ_m = 30°, 32°)					(δ_p, δ_m = 5.0°, 10°)		

NOTES: $N(n)$ = number of cores (specimens) used to calculate the mean direction; D = declination; I = inclination; α_{95} = radius of circle of 95% confidence; κ = Fisher's precision parameter; δ_p, δ_m = semi-axes of oval of 95% confidence.

[a]Magnetite's magnetization is obtained by vector subtraction between 200 and 560°C steps.

[b]Hematite's magnetization isolated after the 640°C heating step by the stable end-point criterion.

[c]Specimens with NNW declination of Fig. 5*c* are not included in this table.

than calculating a mean magnetization carried by hematite in *all* the specimens, the specimen magnetization with NNW declination has been omitted from further discussion. Mean direction of magnetization for the SSW grouping was calculated and a paleomagnetic pole, called the DB2 pole, was calculated using that mean (Table 1).

A baked contact test was conducted of the host rock (Deadman's Bay granite) intruded by dike A14. Specimens of the host rock were drilled starting 2.5 cm away from the dike and extending to 4.5 m. The specimen spacing was at 5 cm intervals close to the dike but increased farther away from the dike. Figure 7 shows the directions of individual specimens of the host rock at NRM level and after AF demagnetization. A separate paleomagnetic study of the Deadman's Bay granite (presently underway) indicated that AF demagnetization in fields of the order of 20 mT is adequate to isolate a characteristic magnetization for that rock unit. There is a remarkable improvement in grouping of the host rock specimens after demagnetization. No change of direction was noticed with distance away from the dike. The mean magnetization obtained in Fig. 7*b* is not dissimilar to the magnetization carried by magnetite in the dikes. Although it could not be confirmed from the contact test whether the magnetization carried by magnetite or hematite in the dikes is primary, lack of shallow inclination magnetization in the host rock specimens close to the dike implies that the hematite's magnetization may not be of primary origin.

All the diabase dikes studied are of the multiple intrusion variety (N. R. Jayasinghe, personal communication, 1982) with diabase intrusion probably taking place in short intervals of time. Although it is tempting to relate the observed NNW declination magnetizations to multiple intrusive nature of the dikes, such correlation implies that the hematite's magnetization is primary. The baked contact test indicates otherwise and thus it is quite unlikely that the observed NNW magnetizations in hematite are related to the multiple intrusions.

An anisotropy test was conducted to check if the directions of magnetization of the dike are systematically deflected away from the direction of the ambient field due to the anisotropic nature of the rock fabric. The procedures adopted are similar to that of Irving and Park (1973). IRM was induced in a field of 865 mT for specimens from dike A14 in the direction of the magnetite's magnetization for that dike. The IRM was progressively demagnetized by alternating fields and Fisher's statistical parameters were calculated. It was found that there was no deflection of the IRM directions away from the direction of the ambient field, the two directions agreeing to within 1° (α_{95} for the grouping of the demagnetized IRM directions is 3.5°). Thus, it was concluded that there was no effect of anisotropy of the

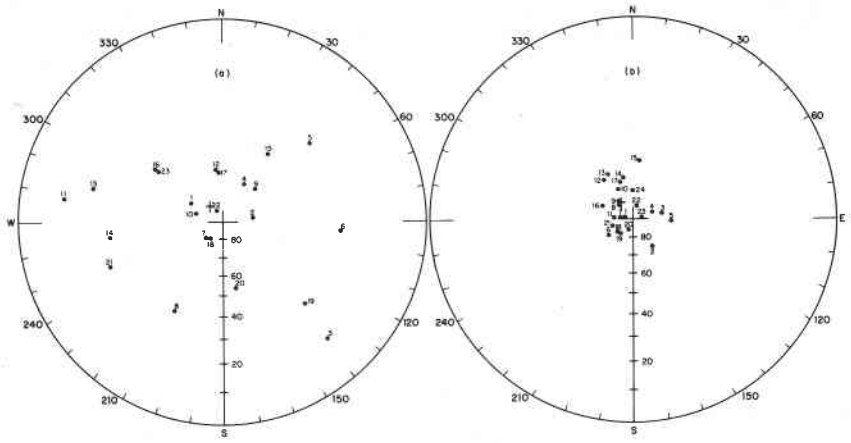


FIG. 7. Results of the baked contact test. Specimen magnetization directions of the host rock (Deadman's Bay granite) at (*a*) the NRM level and (*b*) after demagnetization in fields of the order of 20 mT. Numbers next to the symbols are specimen numbers and increase with distance from the dike. The closest, No. 1, is 2.5 cm, and the farthest, No. 24, is 4.50 m away from the edge of the dike A14.

rock fabric deflecting the magnetization directions away from that of the ambient field.

The dikes thus carry two magnetizations: one by magnetite yielding a pole at 30.1°N, 70.7°W (DB1 pole) and the other by hematite with SSW declinations yielding a pole at 38.6°S, 65.5°W (DB2 pole). From the baked contact test, it can be said that the hematite's magnetization is unlikely to be primary, whereas the magnetization carried by magnetite may be of primary origin, similar to that carried by some granites and other dikes farther east (Murthy 1983). All the available paleomagnetic poles for the Siluro-Devonian age relative to the northern Appalachians are plotted in Fig. 8. The apparent polar wander path for cratonic North America similar to the one compiled by Irving (1979) is also plotted in this figure. The DB2 pole falls close to the Devonian poles for "Acadia." Kent and Opdyke had suggested in a series of papers (Kent and Opdyke 1978, 1979, 1980) that the region of the Maritime Provinces of Canada and the New England portion of the United States is a displaced terrain that was welded to North America only in the Late Carboniferous. The basis for this argument is a "significant" displacement of the Devonian poles from the Maritime – New England region relative to the poles for cratonic North America for the same age. Data presented in Fig. 8 support the displaced terrain hypothesis, as the Devonian poles for Acadia do fall north of the mean Devonian pole for cratonic North America. However, we are hesitant at the moment to calculate a "mean" pole for any part of the Devonian or for the total Devonian relative to Acadia because of the uncertainty of the ages of the magnetizations involved, and the dispersion of poles.

Kent and Opdyke (1979) suggested, in particular, that the displacement of Acadian pole positions from the pole positions for cratonic North America persisted even in Early Carboniferous times. Although a perusal of Fig. 6 of Kent and Opdyke (1979) makes such a difference of pole positions for the Carboniferous likely, the fact that error ovals for individual poles are not plotted in their figure makes it hard to judge such a statement. Unlike the Devonian pole positions and their ages, those of the *Mississippian* are well defined. Mean Mississippian pole positions can be calculated both for cratonic North America and Acadia and an *F*-test can be used to see if these pole positions are significantly different. Such a procedure would be justified as only Mississippian poles are considered and the age difference between the various poles is minimal. The results of such a test are presented in Table 2. A cratonic North American mean pole was calculated using data from St. Joe Limestone (Scott 1979), Mauch Chunk Formation (Knowles and Opdyke 1968), and the Barnett Formation (Kent and Opdyke 1979). An Acadian mean pole was calculated using the poles from Codroy Group, Newfoundland (Nairn *et al.* 1959; Black 1964), Pre-Pictou sandstone (Black 1964), Hopewell Group (Roy and Park 1969), and Shepody and Maringouin Formations (Roy and Park 1974). The results show that at 99% probability the two mean pole positions are not significantly different. Thus, we conclude that whereas the Acadian poles for the Devonian are displaced relative to the cratonic North American poles, such a displacement is not present during Mississippian times.

The DB1 pole lies close to certain "anomalous" pole positions, all due to magnetizations with steep inclinations. Steep inclination magnetizations were noticed in most rock units in the area including the Bonavista Bay

TABLE 2. *F*-ratio test to check for significant difference between mean Mississippian pole positions

	N	λ, ϕ	A_{95} (°)	κ	R	$\frac{(N-2)(R_1+R_2-R)}{N-(R_1+R_2)}$	$F_{2,10}$ at $P = 0.01$
"Acadian" mean pole[a]	4	32.7°S, 53.3°W	9.9°	86.9	3.9655		
Cratonic North America mean pole	3	43.8°S, 57.6°W	10.5°	139	2.9856		
Combined mean Mississippian pole	7	37.4°S, 53.4°W	7.0	74.2	6.9191	3.2720	7.56

NOTES: N = number of poles averaged (see text); λ, ϕ = latitude and longitude of mean pole; A_{95} = radius of the 95% circle of confidence about the mean pole; κ = Fisher's precision parameter; $R(R_1, R_2)$ = length of the resultant vector.

[a]Although arguments can be made for not including the Codroy Group pole from western Newfoundland in the Acadia, it was included in calculating the Acadian mean pole position.

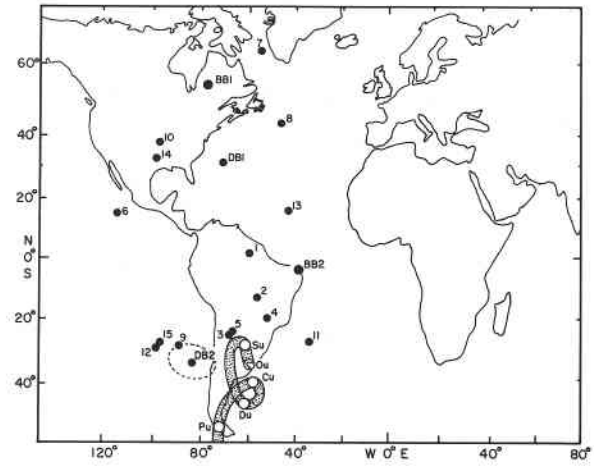


FIG. 8. Siluro-Devonian paleopoles from northern Appalachians (solid circles) with the apparent polar wander path relative to cratonic North America as background. The dotted oval around DB2 represents 95% oval of confidence. (1) Mascarene intrusives;[a] (2) Botwood rhyolite;[b] (3) Botwood sandstone;[b] (4) Hershey Formation;[c] (5) Eastport Formation;[c] (6) Mt. Peyton diorite;[b] (7) Mt. Peyton granite;[b] (8) St. Stephen pluton;[a] (9) St. George pluton;[a] (10) St. George pluton;[a] (11) Clam Bank Group;[d] (12) Traveler felsite;[e] (13) Cape Breton granite;[f] (14) Cape Breton granite;[f] (15) Cape Breton granite;[f] (BB1) Bonavista Bay dikes;[g] (BB2) Bonavista Bay dikes;[g] (DB1) Deadman's Bay dikes (magnetite); (DB2) Deadman's Bay dikes (hematite). [a]Roy *et al.* (1979); [b]Lapointe (1979); [c]Kent and Opdyke (1980); [d]Black (1964); [e]Spariosu and Kent (1980); [f]Rao *et al.* (1981); [g]Murthy (1983).

dikes to the east. The origin of these magnetizations possibly lies in a shearing stress affecting the Gander Zone rocks or as a feature of rock units from the southeastern margin of the Iapetus Ocean (Murthy 1983). Many granite bodies outcrop in the northeast Gander Zone intruded by and intruding the diabase dikes. Paleomagnetic studies of these granites hopefully would reveal the extent of these steep magnetizations.

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